

Day 44 - NAT (pt 1)

Wednesday, September 9, 2026 8:37 PM

NAT Info

Private IPv4 Addresses (RFC 1918)

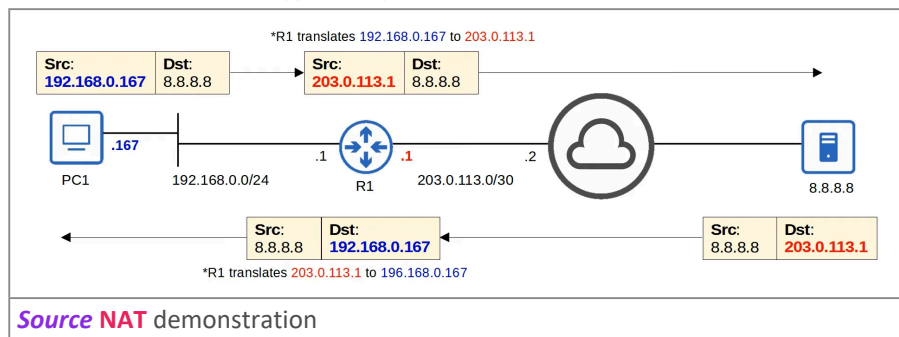
IPv4 doesn't provide enough addresses for all devices that need them in the modern world.

- That's why **RFC 1918** was created as a short term solution for the problem. It specifies three **IPv4** ranges as *private* address ranges (meant for local usage), and they are:
 - 10.0.0.0/8** (10.0.0.0 to 10.255.255.255)
 - 172.16.0.0/12** (172.16.0.0 to 172.31.255.255)
 - 192.168.0.0/16** (192.168.0.0 to 192.168.255.255)
- Private* IP addresses cannot be used over the internet.
- These addresses can be used freely in your local networks. They do not have to be globally unique.

Network Address Translation (NAT) General Information

Network Address Translation (NAT) is a protocol that is used to modify the *source* and/or *destination* IP addresses of packets.

- There are many reasons to use **NAT**, but the most common reason is to allow hosts with *private* IP addresses to communicate with other hosts over the internet.
- Source NAT** is the **NAT** type that you'll have to understand and know how to configure for the CCNA.



Source NAT types, and what will be discussed in this lecture/note

- In this lecture/note, we'll be focusing on **Source Static NAT** and how to set it up.
- Meanwhile, **Dynamic Source NAT** and **Dynamic Source PAT** will be discussed in the next lecture/note. ([Day 45 - NAT \(2\)](#))

NAT Address Types

There are four address types used in NAT, they are:

- Inside Local**
 - The IP address of the *inside* host, from the perspective of the *local* network
- Inside Global**
 - The IP address of the *inside* host, from the perspective of the *global* network
- Outside Local**
 - The IP address of the *outside* host, from the perspective of the *local* network
- Outside Global**
 - The IP address of the *outside* host, from the perspective of the *global* network

Good way to remember the types

- Inside/Outside* refers to the *location of the host*
- Local/Global* refers to the *perspective*

Note on: Outside Local and Outside Global addresses

- For the CCNA, the **Outside Local** and **Outside Global** addresses will be the same and will not change.
- Unless **Destination NAT** is used, the two addresses will be the same. (This is outside the scope of the CCNA certificate)

Static NAT

Static NAT involves **statically** configuring **one-to-one** mappings of **private** IP addresses to **public** IP addresses.

- When traffic from the internal host is sent to the outside network, the router will translate the **source** address.
- Also, this **one-to-one** mapping also allows external hosts to access the internal host via the **Inside Global** address.

Static NAT Configuration

- **show ip nat translations**
 - Shows information about IP addresses that have been translated through **NAT**.
 - Includes the following information:
 - **Protocol** used in the connection
 - **Inside Local** address
 - **Inside Global** address
 - **Outside Local** address
 - **Outside Global** address
- **show ip nat statistics**
 - Shows statistics about the current **NAT** process
 - Includes the following information:
 - Total active translations (**static**, **dynamic**, **extended**)
 - Peak translations (highest amount of translations in one time)
 - **Outside** interfaces
 - **Inside** interfaces
 - Etc.
- **clear ip nat translation ***
 - Clears all **dynamic NAT** translations in the **NAT** translation table
 - (One type of **dynamic** translations is IP address translations with a port included)
- **ip nat inside**
 - Defines the '**inside**' interface(s) connected to the **internal** network
- **ip nat outside**
 - Defines the '**outside**' interface(s) connected to the **external** network
- **ip nat inside source static INSIDE-LOCAL-IP INSIDE-GLOBAL-IP**
 - Configures a one-to-one (**Static NAT**) IP address mapping